23K-2001
BCS-6J

AI Assignment 02

Date:

Answer#01:

a. Variables:

$X_1, X_2, X_3, X_4, X_5, X_6$

Domain of each variable = $\{P, G, E\}$

Unary constraints:

- strong breeze (S) → atleast one corridor must be P
- weak breeze (W) → corridors must contain E & G (no P)
- no breeze → both must be G

Binary constraints:

1. Adjacent corridors cannot both be exits.

   $X_i = E$, then $X_j \neq E$

2. Based on breeze observations:
   - if breeze = S : $(X_i = P) \vee (X_j = P)$
   - if breeze = W : $(X_i, X_j) \in \{(E, G), (G, E)\}$
   - if no breeze : $(X_i = G) \wedge (X_j = G)$

b. Initial domains: $X_i = \{P, G, E\}$

Assign: $X_1 = E$

Applying unary constraints: $X_2 \neq E$, $X_6 \neq E$; (neighbors of X_1)

⇒ domains become: $X_2 = \{P, G\}$, $X_6 = \{P, G\}$

other variables remain: $X_3, X_4, X_5 = \{P, G, E\}$

⇒ Result after forward checking:

$X_1 = E$ $\quad$ $X_3 = \{P, G, E\}$ $\quad$ $X_6 = \{P, G\}$

$X_2 = \{P, G\}$ $\quad$ $X_4 = \{P, G, E\}$

$X_3 = \{P, G, E\}$ $\quad$ $X_5 = \{P, G, E\}$

Date:

c. MRV selection

Rule: choose variable with minimum remaining values.

Variable	Domain size
X_1	1
X_2	2
X_3	3
X_4	3
X_5	3
X_6	2

Thus, MRV may select X_2 or X_6. (domain = 2)

d. Assuming $X_6 = G$: $(X_1, X_2, X_3, X_4, X_5, X_6)$

→ Adjacent to X_6 must be consistent.

→ Possible assignments must satisfy

- no adjacent exits
- breeze constraints
- domain constraints

⇒ Possible solutions:

i. (E, G, P, G, E, G)

ii. (E, P, G, P, G, G)

e. MRV again with $X_6 = G$

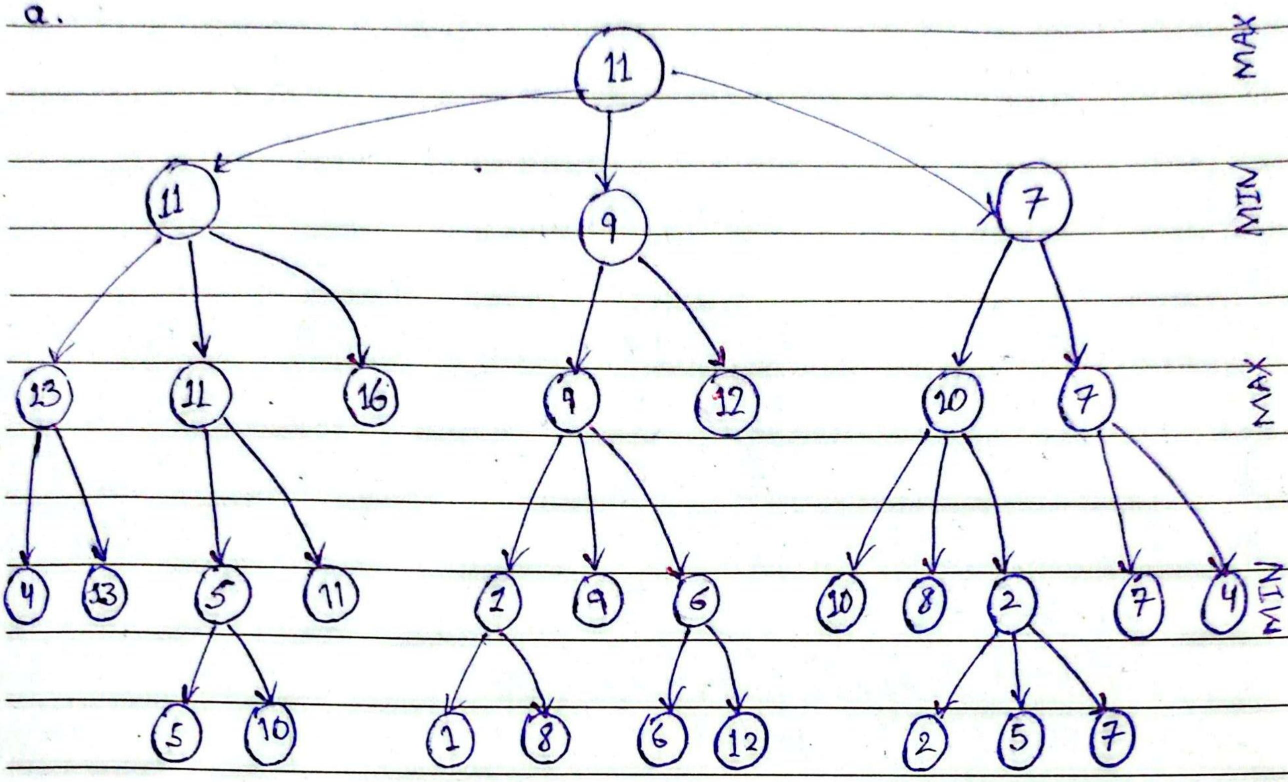
→ Reduce neighbor domains.

Thus, MRV selects:

X_2 or X_5

Answer#02:

a.



b. Alpha-beta Pruning:

Root(max): $\alpha = -\infty$, $\beta = \infty$

left min:

→ $\beta = 13$

→ $\beta = \min(13, 11) = 11$

→ $\beta = \min(11, 16) = 11$

root updates: $\alpha = \max(-\infty, 11) = 11$

Middle min: $\alpha = 11$, $\beta = \infty$

→ $\beta = 9$

→ $\because \beta \leq \alpha \rightarrow 9 \leq 11$

prune branch (12)

→ return 9

Right min: $\alpha = 11$, $\beta = \infty$

→ $\beta = 10$

→ $\because 10 \leq 11$

prune subtree (7, 4)

→ return 7

⇒ Final root value: 11 Ans.

Pruned branches:

- middle subtree leaf (12)
- right subtree branch (7, 4)

Date:

c. No.

Alpha-Beta Pruning only removes branches that cannot affect the decision. The final value at the root always remains the same as mini-max

Answer#03:

Document/Word	I	Am	We	Are	Groot	Total
1	12	12	1	1	13	39
2	17	17	0	0	17	51
3	14	14	2	2	15	47

a) $\frac{1}{3}\left(\frac{13}{39} + \frac{17}{51} + \frac{15}{47}\right) \approx 0.3286$

b) $\frac{1/39}{1/39 + 0/51 + 2/47} = 0.3760$

c) $\frac{17/51}{13/39 + 17/51 + 16/47} \approx 0.3310$

d) $\frac{1}{6} \cdot \frac{13}{39} + \frac{1}{3} \cdot \frac{17}{51} + \frac{1}{2} \cdot \frac{15}{47} \approx 0.3262$

e) $\frac{1/6 \,.\, 1/39}{(1/6 \,.\, 1/39) + (1/3)(0/51) + (1/2 \,.\, 2/47)} \approx 0.1673$

f) $\frac{1/3 \,.\, 17/51}{1/6 \,.\, 13/39 + 1/3 \,.\, 17/51 + (1/2 \,.\, 16/47)} \approx 0.3298$

Date:

Answer#04:

a) $P(R) = P(R|S) * P(S) + P(R|\neg S) * P(\neg S)$

$= 0.7 * 0.25 + 0.3 * 0.75 = 0.4$

b) $P(We) = P(We|R, W_o) * P(W_o|R) * P(R) + P(We|R, \neg W_o) *$
$(1 - P(W_o|R)) * P(R) + P(We|\neg R, W_o) * P(W_o|\neg R) *$
$(1 - P(R)) + P(We|\neg R, \neg W_o) * (1 - P(W_o|\neg R)) *$
$(1 - P(R))$

$= 0.12 * 0.7 * 0.4 + 0.25 * 0.3 * 0.4 +$
$0.1 * 0.2 * 0.6 + 0.08 * 0.8 * 0.6 = 0.114$

c) $P(M|S)$: This value is useful first $\{P(We|S) = P(We|R, W_o) *$
$P(W_o|R) * P(R|S) + P(We|R, \neg W_o) * (1 - P(W_o|R)) *$
$P(R|S) + P(We|\neg R, W_o) * P(W_o|\neg R) * (1 - P(R|S)) +$
$P(We|\neg R, \neg W_o) * (1 - P(W_o|\neg R)) * (1 - P(R|S))$.

$= 0.12 * 0.7 * 0.7 + 0.25 * 0.3 * 0.7 + 0.1 * 0.2 * 0.3$
$+ 0.08 * 0.8 * 0.3 = 0.1365$

; $P(M|S) = \dfrac{P(M, S)}{P(S)}$

$= \dfrac{P(M|We) P(We|S) P(S) + P(M|\neg We)(1 - P(We|S)) P(S)}{P(S)}$

$= P(M|We) * P(We|S) + P(M|\neg We) * (1 - P(We|S))$

$= 0.02 * 0.1365 + 0.42 * 0.8635$

$= 0.3654$

Date:

d) $P(B|S)$: This value is useful first $\{P(W_0|S)$

$= P(W_0|R) * P(R|S) + P(W_0|\neg R) * (1 - P(R|S))$

$= 0.7 * 0.7 + 0.2 * 0.3$

$= 0.55\}$;

$P(B|S) = P(B|S, W_0) * P(W_0|S) + P(B|S, \neg W_0) * (1 - P(W_0|S))$

$= 0.8 * 0.55 + 0.4 * 0.45$

$= 0.62$

e) $P(S|B, \neg W_0)$: This value is useful first $\{P(W_0|\neg S)$

$= P(W_0|R) * P(R|\neg S) + P(W_0|\neg R) * (1 - P(R|\neg S))$

$= 0.7 * 0.3 + 0.2 * 0.7$

$= 0.35\}$;

$$P(S|B, \neg W_0) = \frac{P(B, S, \neg W_0)}{P(B, \neg W_0)}$$

$$= \frac{P(S, \neg W_0)[(1 - P(R))(P(S))P(S) + (1 - P(\neg R))(1 - P(S))P(S)}{P(S, \neg W_0)(1 - P(S))P(S) + P(\neg S, \neg W_0)(1 - P(\neg S))P(\neg S)}$$

$$= \frac{0.4[(1 - 0.7)0.7 * 0.25 + (1 - 0.2)(1 - 0.7) * 0.25]}{0.4(1 - 0.55)0.25 + 0.4(1 - 0.35)0.75}$$

$$= \frac{0.045}{0.24}$$

$= 0.1875$

Date:

Answer#05:

a) It is worse for regular emails to be classified as spam than it is for spam emails to be classified as regular email.

$P(study|spam) = 1/10$

$P(study|regular) = 2/3$

$P(free|spam) = 9/10$

$P(free|regular) = 1/3$

$P(money|spam) = 1/2$

$P(money|regular) = 1/3$

b) $p(spam)P(study|spam)(1-P(free|spam))(P(money|spam))$

$= p(spam) * 1/200 (1-p(spam)) P(study|regular)(1-P(free/regular))^{*} (P(money|regular))$

$= (1-p(spam))^{*} 4/27 \; P(spam|s)$

$= p(spam)/200 \,/\, [p(spam)/200 + (1-p(spam))^{*} 4/27]$

$= p(spam)/200 \,/\, [4/27 - p(spam) * 773/5400]$

$= 0.003736$

c) $0.5 = p(spam)/200 \,/\, [4/27 - p(spam) * 773/5400]$

iff $4/27 - p(spam) * 773/5400$

$= p(spam)/100$ iff $p(spam) = 0.9674$

Date:

Answer#06 :

Step 1: Decide which variable is the independent variable and which is the dependent variable:
In this example, the number of crabs is the independent variable (x), and the number of clams remaining is the dependent variable (y). This is because the researchers chose the number of crabs to place in each plot, and the resulting number of clams depends on how many crabs are present.

Step 2: Calculate x^2 for every value of x, and y^2 for every value of y. For each pair of values, calculate $x \times y$.
You should end up with ten values (the same number of values of x) for x^2, ten values of y^2, and ten values for $x \times y$.

x^2	y^2	$x \times y$
4	18769	274
16	4900	280
4	33856	368
25	0	0
16	1225	140
0	88209	0
9	14884	366
25	1	5
1	64009	253
9	22500	450

Date:

Step 3: Calculate Σx, Σy, $\Sigma(x)^2$, $\Sigma(y)^2$, $\Sigma(x \times y)$

$\Sigma x = 29$ $\Sigma((y)^2) = 248353$

$\Sigma y = 1249$ $\Sigma((x)^2) = 109$

$\Sigma(x \times y) = 2136$

Step 4: equation of a line: $y = mx + b$

Step 5: calculate m (slope): $\dfrac{n(\Sigma(x \times y)) - \Sigma x \Sigma y}{n(\Sigma(x^2)) - \Sigma(x)^2}$

$$m = \frac{(10)(2136) - (29)(1249)}{(10)(109) - (29)^2}$$

$$m = \frac{-14861}{249}$$

$$m = -59.683$$

Step 6: calculate b (intercept): $\dfrac{\Sigma y - m(\Sigma x)}{n}$

$$b = \frac{1249 - (-59.683)(29)}{10}$$

$$b = 297.98$$

Step 7: $y = mx + b$

$\Rightarrow y = -59.683x + 297.98$ Ans.

23K-2001 BCS-6J